HAPTION
Atelier Relais
ZA – Route de Laval
53210 Soulge/Ouette
France
http://www.haption.com
contact@haption.com

HAP/05/RT.005/Rév. 4.40

January 08, 2020

USER MANUAL
VIRTUOSEDEMO™ 4.40

Arnaud Geneslay and Mélanie Lelaure

TABLE OF CONTENTS

1 INTRODUCTION

This document is the user manual of the application called “VirtuoseDemo”, version 4.40. It demonstrates the major functionalities of the Virtuose API.

To date, the application runs on Microsoft Windows© and Linux™.

2 INSTALLATION

The installation consists in copying the contents of the folder named after your operating system (win32, linux) to your hard disk.

3 STARTING THE APPLICATION

3.1 UNDER MICROSOFT WINDOWS©

Double-click on the file "start_demo.bat” situated in the folder “Windows/win32”.

A DOS window appears showing the starting process.

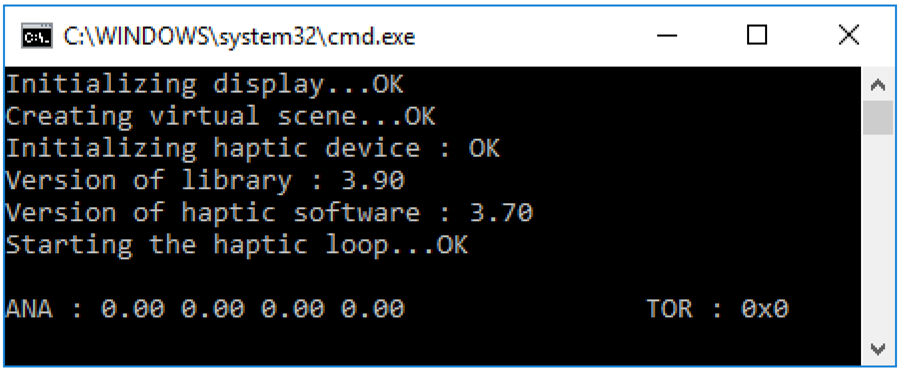


Figure 1

In case of a message "Communication Failure’, check the IP address of the Virtuose™ controller by editing the file “start_demo.bat” (right-click on the file, then choose “modify”).

3.2 UNDER LINUX™

In a shell (for csh-like shells, the syntax can vary somewhat), calibrate the device if it is not already done:

```
[root@usr home]# sudo ./calibrate_<xxx>
```

Launch the deamon of your device to run the haptic service:

```
[root@usr home]# sudo ./SvcHaptic_<xxx>
```

Check the log file, if there is no error:

```
[root@usr home]# cat /var/log/SvcHaptic_<xxx>
Starting SvcHaptic service
Deamon started successfully
```

"cd" to the folder containing the file "virtuoseDemo”.

Copy the virtuoseAPI.so library given with your device next to the virtuoseDemon binary file.

```
[root@usr home]# cp <path_to_virtuoseAPI.so>/virtuoseAPI.so  .
```

Enter the following commands to launch the virtuoseDemo:

[root@usr home]# **export LD_LIBRARY_PATH=.**

[root@usr home]#**./virtuoseDemo -mv [controller IPaddress #portToApi] [Virtuose type] [virtual object mass]**

```
VirtuoseAPI version: 4.01
Initializing display...OK
Creating virtual scene...OK
Initializing haptic device : OK
Device type : virtuose 6D
Serial number : 130
Version of library : 4.01
Version of haptic software : 6.04
Starting the haptic loop...OK
```

The `start_demo.sh` script launches the last two commands, check the virtuoseDemo parameters before using it.

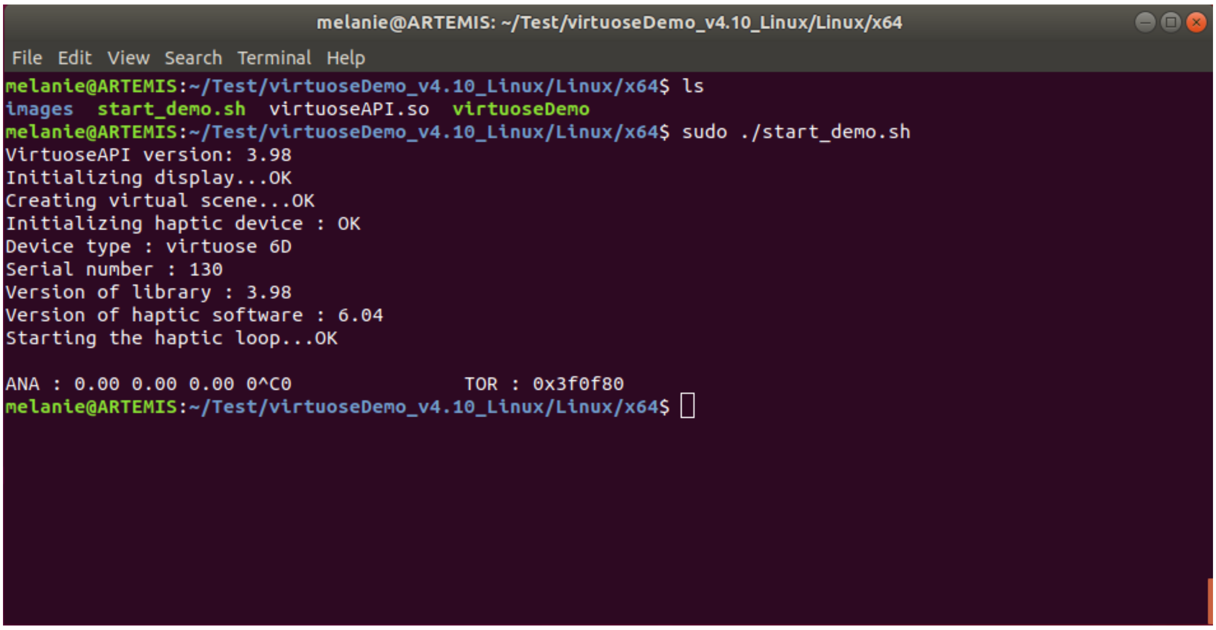


Figure 2

In case of a "`Communication Failure`" message, check the IP address and the portToApi of the Virtuose™ controller. For some devices, you might need to run the demo in `sudo` mode.

The message `VirtuoseAPI version:` **`0.0`**, means that the virtuoseAPI dynamic library has not been correctly loaded. Please check that the virtuoseAPI.so has been correctly copied in the virtuoseDemo folder and run again the **`export LD_LIBRARY_PATH=.`** command.

The parameters of the ./virtuoseDemo are :

controller IPaddress : IP of the PC running the `SvcHaptic_<xxx>` `deamon`. If the deamon is launch locally, on the same PC as the virtuoseDemo, the ip is **`127.0.0.1`**

portToApi: The value of this port can be found in the `/etc/Haption/Svc_Haptic_<xxx>.conf` file. The default value is either **`5000`** or **`53210`**, depending on the devices.

virtuose type: The `start_demo.sh` script lists all the types. For a standard Virtuose6D it will be **`v6dv4`**.

virtual object mass: The default value is **`1.0`** for 1kg.

4 USING THE APPLICATION

The demo application is composed of several windows. The viewing window shows the test scene (*Figure 3*).

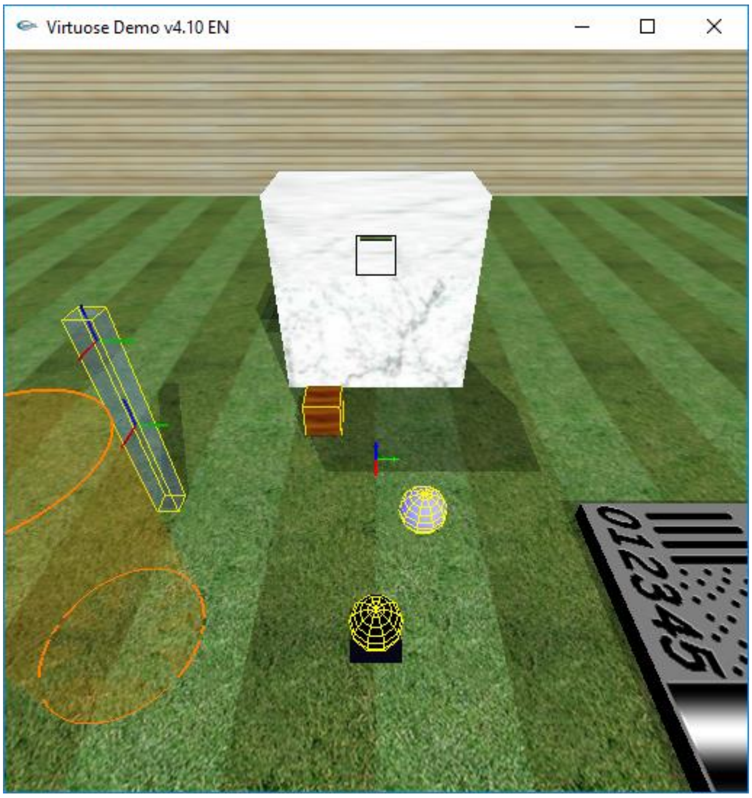


Figure 3

Another viewing scene is specifically design for the HGlove™ (*Figure 4*).

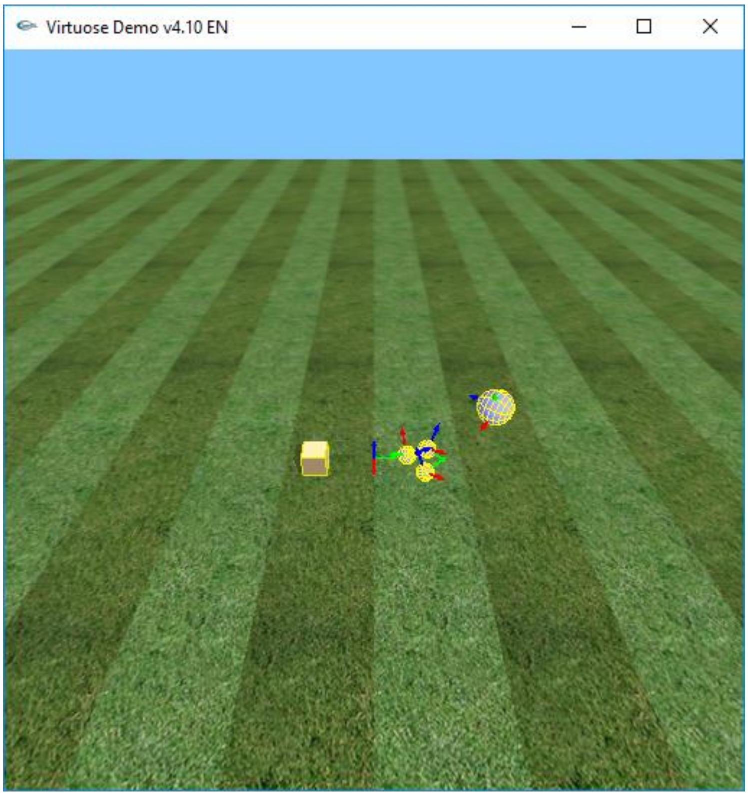


Figure 4

A second window, called Control Panel (*Figure 5*), shows the current state and lets you activate the functionalities of the Virtuose. One Control Panel appears for each connected Virtuose haptic device.

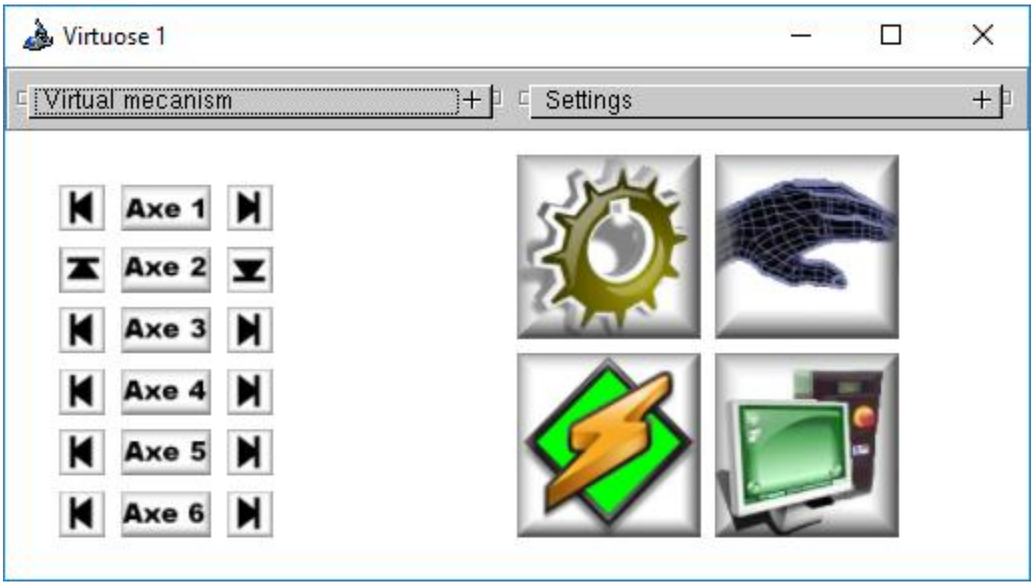


Figure 5

For the HGlove™, there is one panel for each finger.

4.1 USE OF THE VIRTUOSE™ PUSH-BUTTONS

The first push-button activates and deactivates a virtual mechanism after it has been selected on the Control Panel.

The second push-button is used for offset. When pushed, the position of the reference frame attached to the end-effector of the Virtuose™ is frozen.

The third push-button attached the Virtuose™ to avatar closest virtual object.

4.1.1 COBRA Handle

Figure 6

4.1.2 TREH Handle and Virtuose TAO

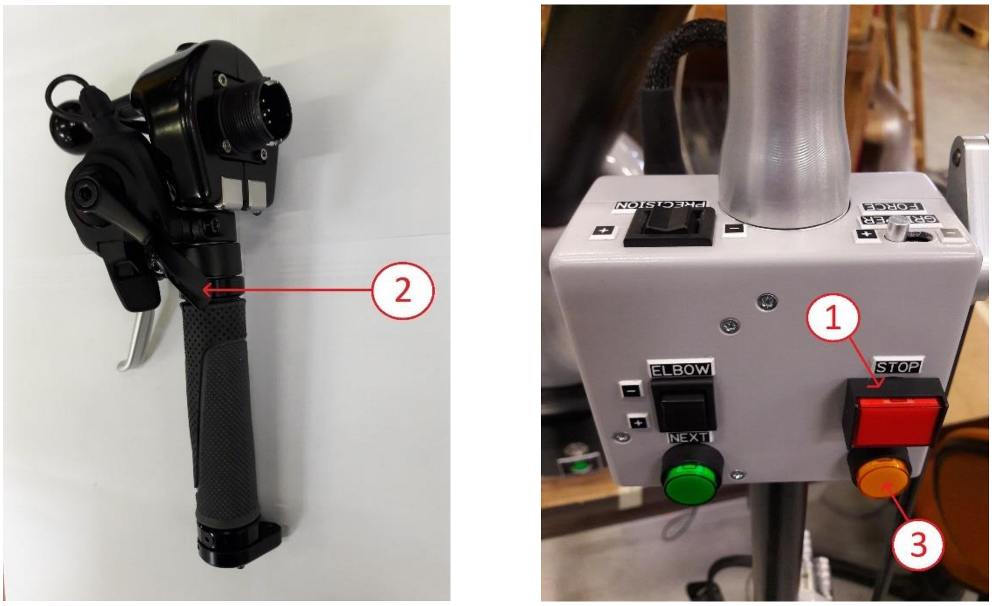


Figure 7

4.2 VIEWING WINDOW

When connected to one Virtuose, a reference frame representing the position of the end-effector appears in the scene, in the form of three vectors (red for X, green for Y, and blue for Z). The object closest to its position is highlighted orange, in order to show that it can be attached to.

To attached on object to the Virtuose press the third push-button or the keyboard space bar.

When an object is attached to the Virtuose, it stays highlighted orange. When it is in collision with other objects in the scene, it turns red.

When moving an object, you will probably reach the end of the arm workspace sooner or later. Use the second push-button to generate an offset and bring the arm back to the center of its workspace.

4.2.1 Navigation with the mouse

Rotate the viewpoint around axes X and Z of the camera by pushing the left mouse button. The pointer (a) appears during the operation (see *Figure 8*).

Translate the viewpoint inside the plane XY by pushing the right mouse button. The pointer (b) appears during the operation.

By pushing simultaneously, the right and left mouse buttons, you can translate the viewpoint inside the XZ plane (pointer (c) appears).

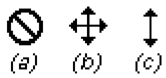


Figure 8

4.2.2 Navigation with the Virtuose

The modification of the viewpoint using the mouse mode of the Virtuose is exactly the same as explained above. In order to activate the mouse mode, double click on the first push-button of the Virtuose.

4.3 CONTROL PANEL

Using the Control Panel, you can change the operating mode of the Virtuose™, activate bilateral constraints (also called “virtual mechanisms”), modify scale values and force saturation.

The window contains three components:

- A set of icons showing the status of the device.
- A pull-down panel dedicated to virtual mechanisms.
- A pull-down panel dedicated to control parameter.

4.3.1 Device status

A set of icons shows the current status of the Virtuose device.

4.3.1.1 Workspace bounds

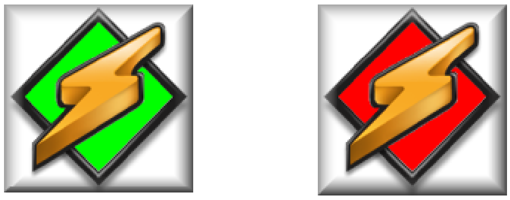
In case the Virtuose™ hits one of its workspace bounds, the corresponding icon turns red (*Figure 9b*). You can come back inside the workspace by pushing the indexing (or offset, second button) button:

(a) (b)

Figure 9

4.3.1.2 Force-feedback activation

When the force-feedback is activated, the power icon turns red (*Figure 10*).



(a) force-feedback active *(b) force-feedback inactive*

Figure 10

4.3.1.3 Connection

When the connection with the Virtuose™ is active, the icon is green (*Figure 11*).

(a) not connected *(b) connected*

Figure 11

4.3.1.4 Virtual Guide

When a virtual guide is active, the corresponding icon turns green. In order to activate a virtual guide, you must first select it on the pull-down panel, and then press the first push-button (*Figure 12*).

Figure 12

4.3.1.5 Attachment

When an object is attached to the Virtuose, the corresponding icon is highlighted. In order to attach to an object, you must first move the Virtuose™ frame near it, and then press the third push-button (*Figure 13*) or SPACE key.

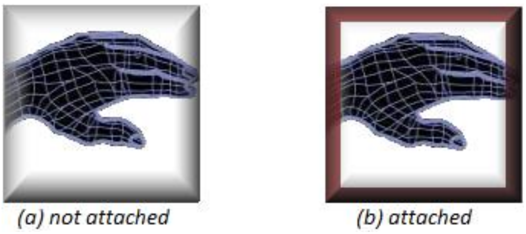


Figure 13

4.3.1.6 TOR inputs

In the DOS windows, the field TOR give the value of the inputs of the arm in hexadecimal.
This value is a field of bits. For a TAO arm, the bits are the following value:

Bit n°0 : force feedback
Bit n°1: dead man
Bit n°2: shift
Bit n°3: gripper lock
Bit n°4 : next
Bit n°5 : rotation +
Bit n°6 : rotation –
Bit n°7 : stop
Bit n°8 : wait
Bit n°9 : redundancy + (or Z-Elec +)
Bit n°10 : redundancy - (or Z-Elec -)
Bit n°11 : precision +
Bit n°12 : precision –
Bit n°13 : gripper force +
Bit n°14 : gripper force –
Bit n°15 : calibration

4.3.1.7 Analogic input

In the DOS windows, for a TAO arm, the field ANA indicates the position of the gripper between 0.0 and 1.0. This value corresponds to a closing percentage.

For the other arm, this value is 0.0.

4.3.1.8 The gripper

In the DOS windows, for a TAO arm, the field GRIPPER indicates the position of the gripper between -0.5 and 1.5 taking the lock into account. The value of the gripper can be locked or unlocked by using the switch GRIPPER on the TREH handle.

4.3.2 Virtual mechanism

The pull-down panel "Virtual mecanism" gives access to several options of the Virtuose™ control mode (*Figure 14*). The activation and deactivation of the virtual mechanisms are done by pressing the first push-button.

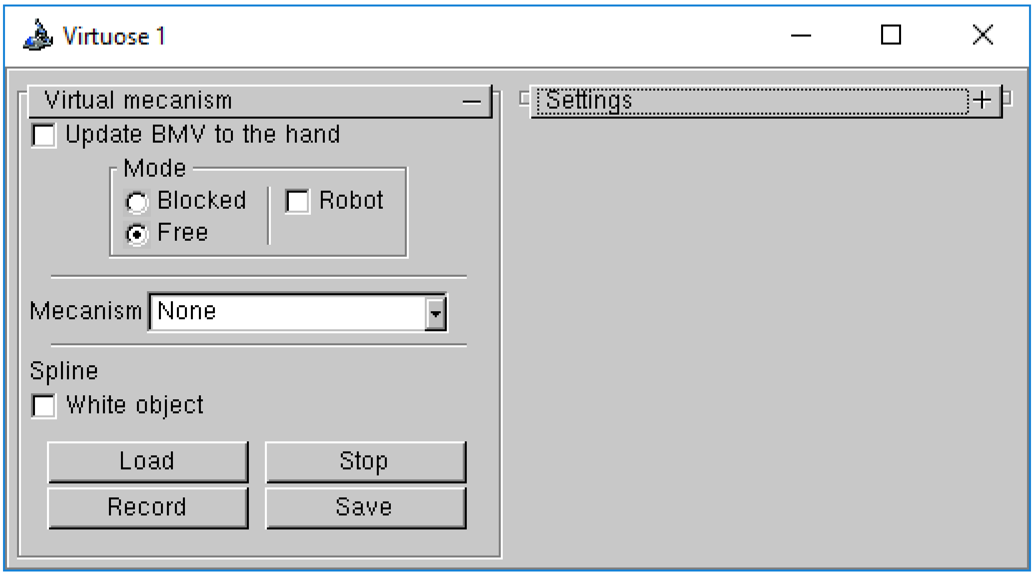


Figure 14

- "Update BMV to the hand": The base of the virtual mechanism is set at the current position of the end-effector.
- "Mode Bloqued/Free": The default control mode is either fixed (when the current virtual mechanism is deactivated, the Virtuose™ cannot move), or free (on the contrary, it can move freely).
- "Mode Robot": Starts an automatic movement along a spline or line segment.
- "Mecanism" pull-down menu:
  - "Cartesian displacement": Line segment between the position of the base reference frame and the current position of the end-effector.
  - "Tx": The end-effector can only move in translation along axis X of the virtual mechanism base frame.

- "Rx": The end-effector can only move in rotation around axis X of the virtual mechanism base frame.
- "TxTy": The end-effector can only move in translation inside the XY plane of the virtual mechanism base frame.
- "TxRx": The end-effector can only move in translation and rotation along axis X of the virtual mechanism base frame.
- "TxTyTz": The end-effector can only move in translation.
- "RxRyRz": The end-effector can only move in rotation around the center of the virtual mechanism reference frame.
- "Manivelle": Same as "RxTx".

- "Spline":
  - "White object": Inhibit the visual textures of other objects, in order to see the spline virtual mechanism better.
  - "Load": Load a spline virtual mechanism.
  - "Record": Start recording the movements of the VirtuoseTM device, in order to create a spline virtual mechanism.
  - "Stop": Stop recording movements and show the spline on the screen.
  - "Save": Save the recorded spline to a file for later use.

After the spline is shown on the screen, you can start using it: press the first push-button to activate a line segment going to the first point of the spline; when you're there, the spline is automatically activated.

4.3.3 Control parameters

The pull-down panel "Settings" gives access to several control parameters (*Figure 15*).

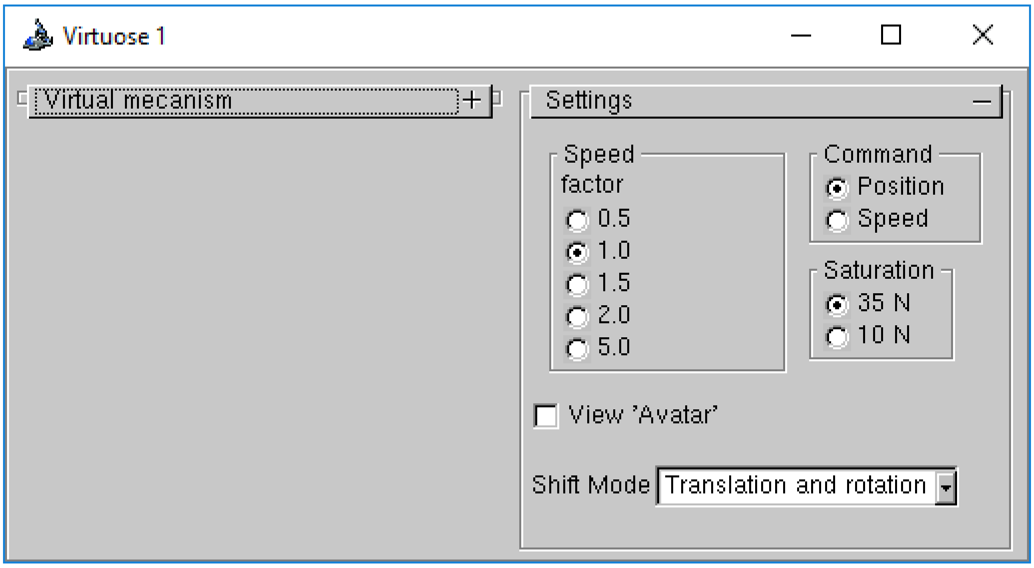


Figure 15

- "Speed factor" modifies the displacement scale factor between the physical workspace of the Virtuose™ and the virtual space of the scene.
- "Command":
  - "Position": default control mode.
  - "Speed": when leaving the volume of a sphere centered at the center of the device workspace, the movements of the objects are speed-controlled.
- "Saturation": Change the maximum force.
- "View ‘Avatar’": Let the camera follow the end-effector of the Virtuose™ device.